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Smart Waste Collection DT

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Timeline

1 INTRODUCTION

2 ARCHITECTURE AND REPORTING

3 DEMONSTRATION

4 CONCLUSION

1

INTRODUCTION

1. Context

🧾 Facts:

💥 Disruptions in waste collection due to weather and reduced pickup frequency. (CityNews, 2025)

🗑️ Overflowing bins and litter impacting city cleanliness and public health. (Reddit, 2025)

Controversy over landfill expansion to accept U.S. hazardous waste.

👨 Calls for better waste management policies and improved city services. (Ensemble

💻 Montréal, 2025)

Introduction | Architecture/Report | Demo | Conclusion

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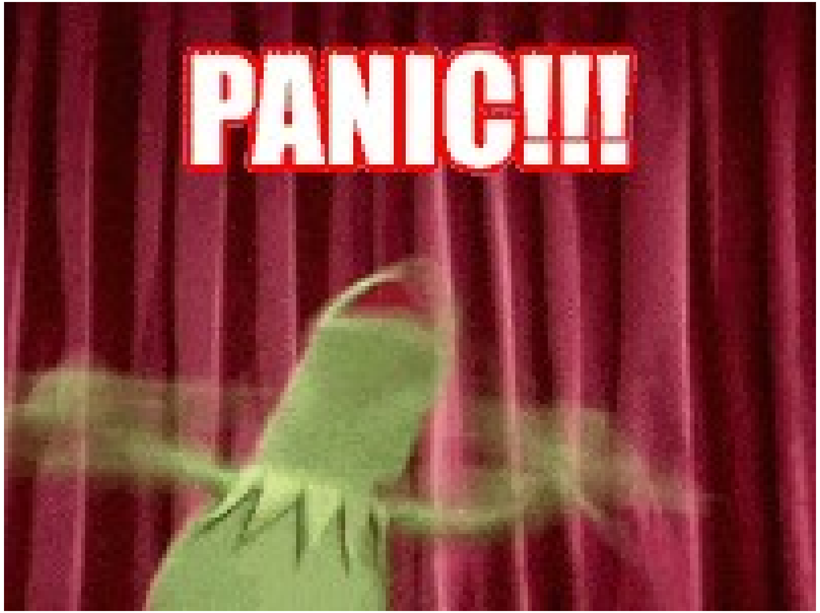
But don't worry!



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WasteTwin is here to the rescue!

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We consider the smart waste collection system a **System of Systems (SoS)**

❗ Conditions for a System to be a SoS :

- 👨 Managerial Independence. (Maier, 1998).
- ⚙️ Operational Independence. (Maier, 1998).
- 🌎 Geographic distribution. (Andrew and Christopher, 2001)
- ♾️ Evolutionary Development. (Andrew and Christopher, 2001)
- 🌆 Emergent Behavior. (Andrew and Christopher, 2001)

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Our SoS has three **Constituent Systems (CSs)**

CS1 - Traffic Lights

Operation: turn on/off lights

Our SoS has three **Constituent Systems (CSs)**

CS1 - Traffic Lights

CS2 - Waste Trucks

Operation: Circulate in the city

Our SoS has three **Constituent Systems (CSs)**

Operation: Fill Level

4. What has Been Done

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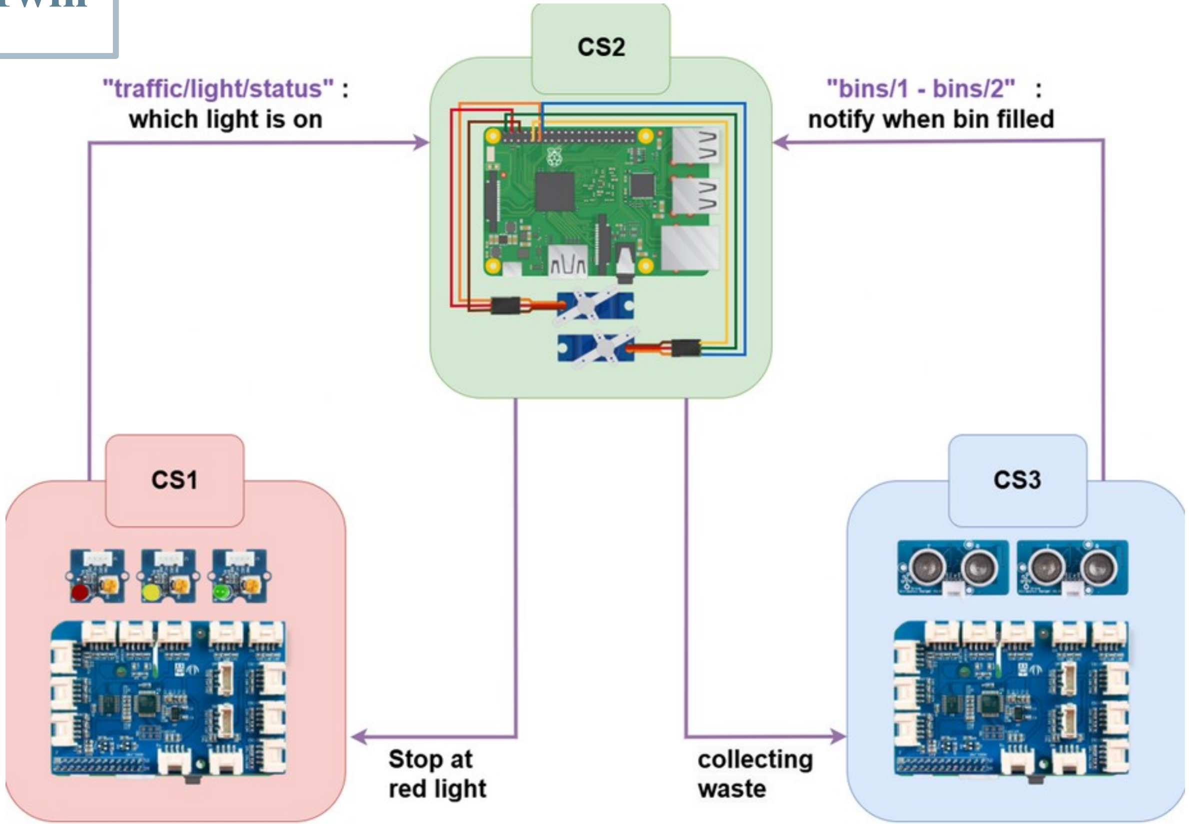
📈 Progress Achieved :

- 📡 Create the physical system using IoT sensors.
- 🚛 Realise a Digital Shadow of the trucks.
- 🚦 Realise a Digital Shadow of the traffic lights.
- 🗑️ Realise Digital Twin of the waste bins.

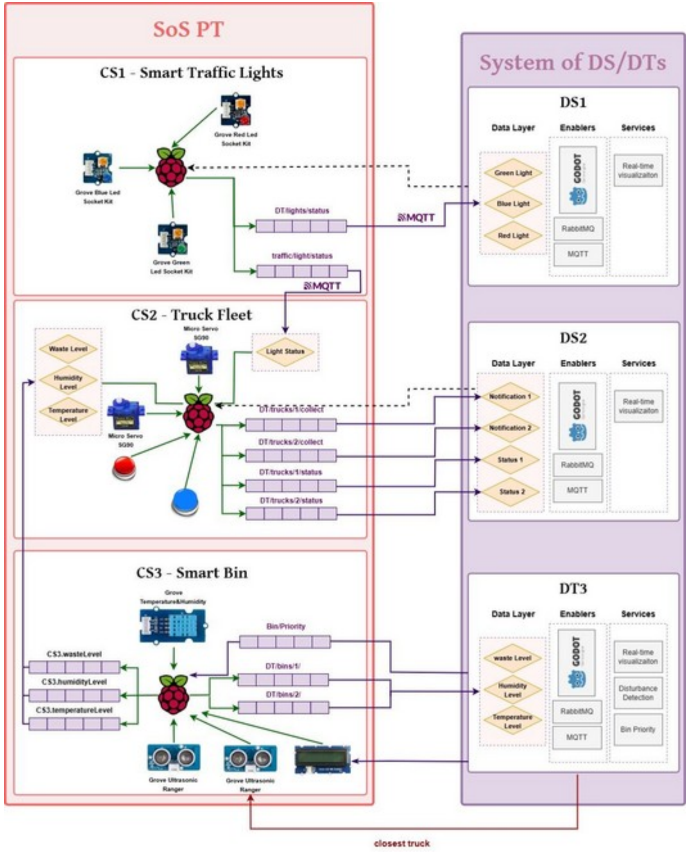
2 ARCHITECTURE AND REPORTING

1. Physical Twin



MC1 - System under study

2. Architecture



CS : Constituent System
SoS : System of Systems
DT: Digital Twin
DS: Digital Shadow

2. Architecture

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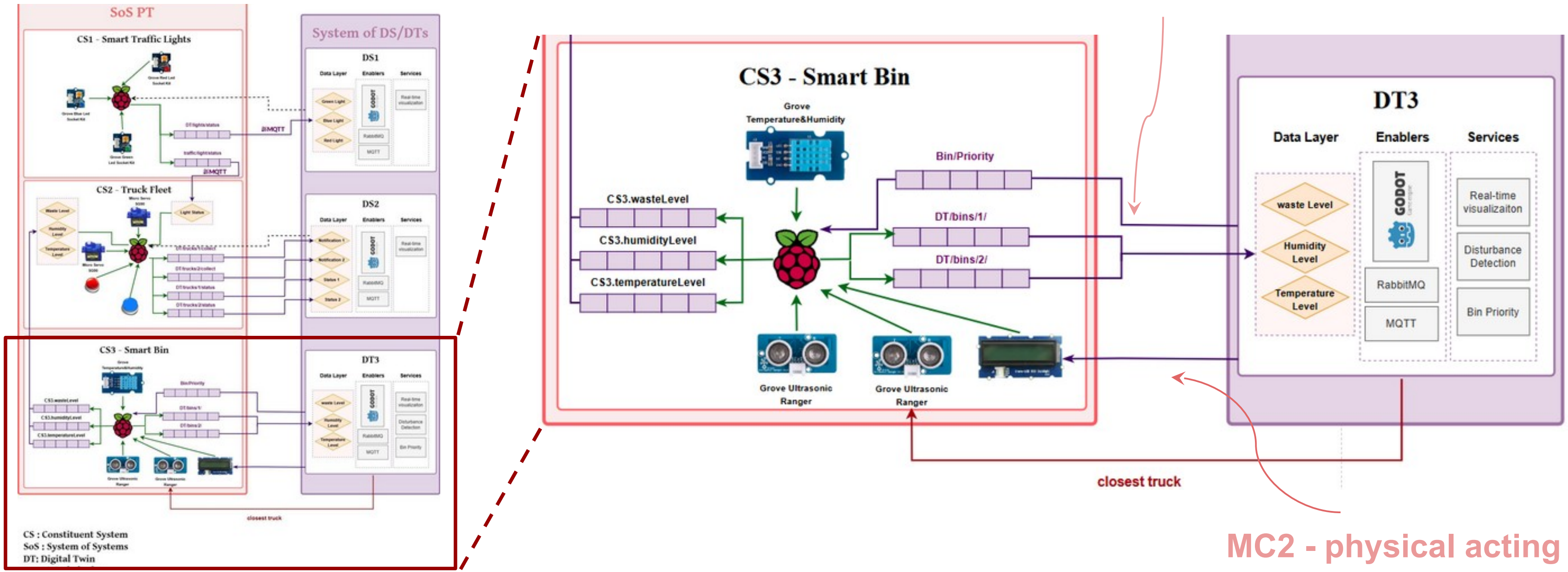
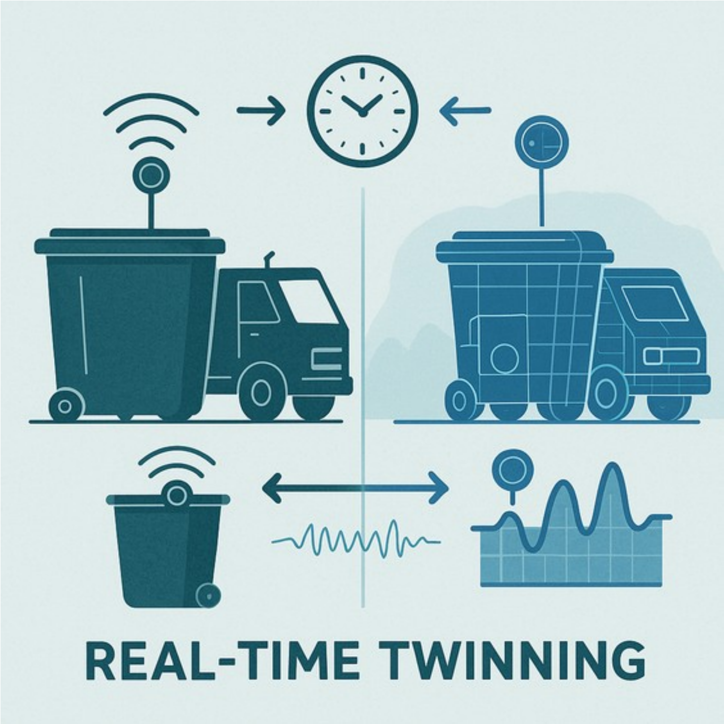
MC3 - physical sensing component



MC2 - physical acting component



MC7: Twinning Timescal :

The data is sent from and to the digital twin in real-time

MC9: Lifecycle :
Operation phase

3 DEMONSTRATION

1. Used Tools

GODOT
Game engine

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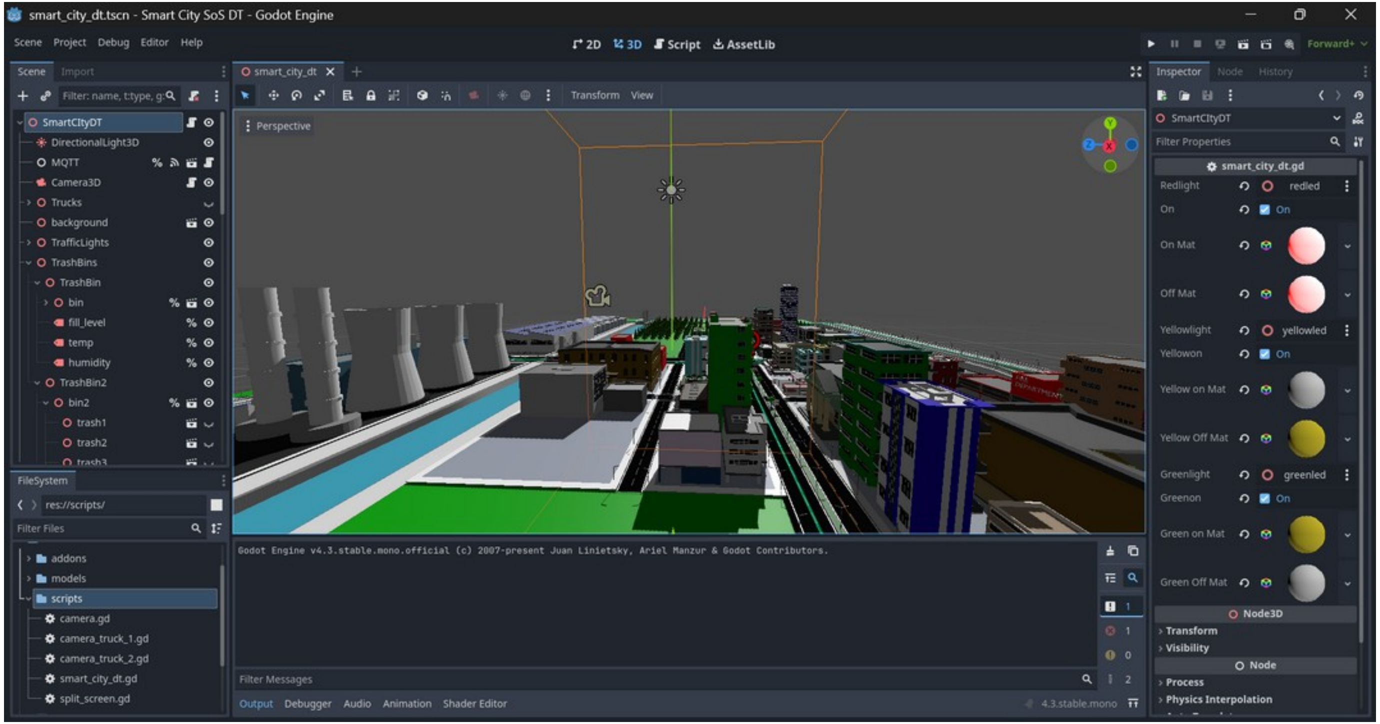

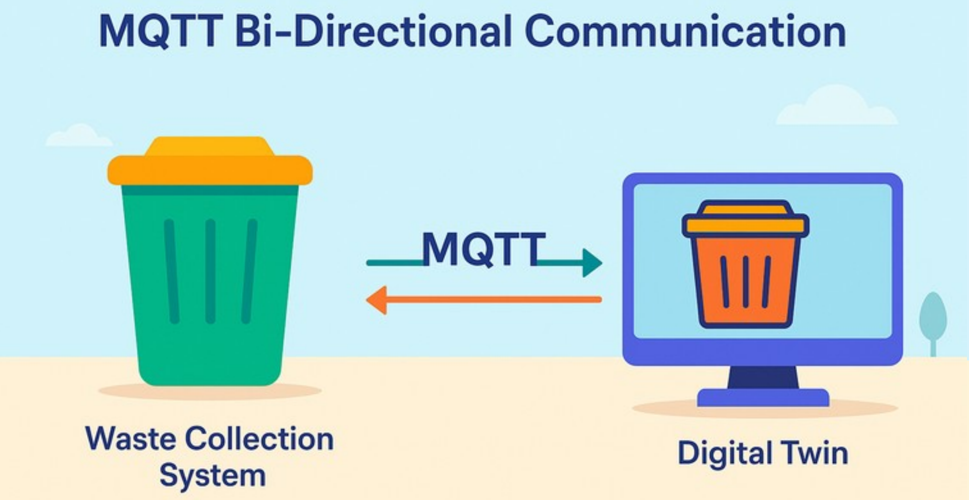
MQTT Bi-Directional Communication
MQTT
Waste Collection System
Digital Twin

3. Coding Languages

4. Followed Steps to create the DT

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1. Create a bin model in Blender
2. Use the .glb file in Godot

3. Sending the data from the physical to the digital asset making it a Digital Shadow

4. Simulate a disturbance (exemple: rain) that the DT would detect and inform the PT

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And now for the fun part …

4 CONCLUSION


Thanks to WasteTwin
the city manages waste and
collections better

REFERENCES

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(The Guardian,2025) https://www.theguardian.com/us-news/2025/mar/06/canadians-toxic-waste-trump-tariffs

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ChatGPT was used for some image generation